

SF1811 Optimization

Lecture 1: Introduction to optimization and linear programming

Chapters 1–3

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Introduction to optimization and linear programming

linear optimization

Main question:

How can a practical decision problem be formulated as a linear programming problem?

(LP)

Today:

- Optimization problems.
- The Tables and chairs problem.
- Geometric interpretation.
- Linear programming in standard form.

Optimization problem

Formulate → Real application

Solve → x^* solution

For us: $x \in \mathbb{R}^n$

Basic form:

$$\begin{aligned} \text{minimize} \quad & f(x) \\ \text{subject to} \quad & x \in F. \end{aligned}$$

- x: decision variables,
- f: objective function,
- F: feasible region.

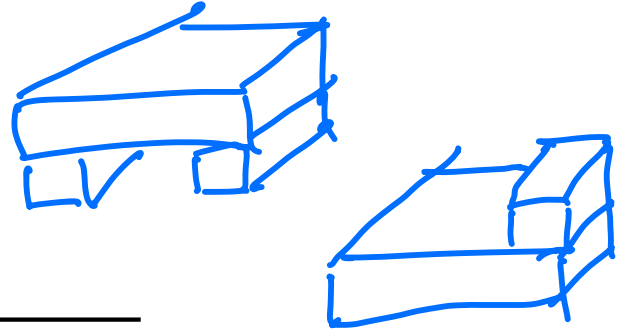
Optimality certificate

A point $x \in F$ is a feasible solution.

An optimal solution is a feasible solution with the best objective value.

$$\underset{x \in F}{\text{maximize}} \; f(x) \iff \underset{x \in F}{\text{minimize}} \; -f(x).$$

Tables and chairs



A manufacturer produces tables and chairs.

	Large parts	Small parts	Revenue
One table	1	2	SEK 400
One chair	1	1	SEK 300

Available resources:

- 200 large parts,
- 300 small parts.

How many tables? T

How many chairs? C

How many tables and chairs should be produced to maximize total revenue?

Formulation of Tables and chairs

Introduce the decision variables

$$T = \text{number of tables}, \qquad C = \text{number of chairs}.$$

- Objective function: $\max 400T + 300C \iff \min -400T - 300C$
- Resource constraints: $1 \cdot T + 1 \cdot C \leq 200$ Big parts
  $3 \cdot T + 1 \cdot C \leq 300$ Small parts
- Sign restrictions: $T \geq 0$, $C \geq 0$

We assume T and C can take on real values

Geometric interpretation

$\begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} T \\ C \end{pmatrix} \leq 200$

Constraints:

$$T + C \leq 200,$$
$$2T + C \leq 300,$$
$$T, C \geq 0.$$

$T \geq 0 \quad \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} T \\ C \end{pmatrix} \geq 0$

$\begin{pmatrix} T \\ C \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

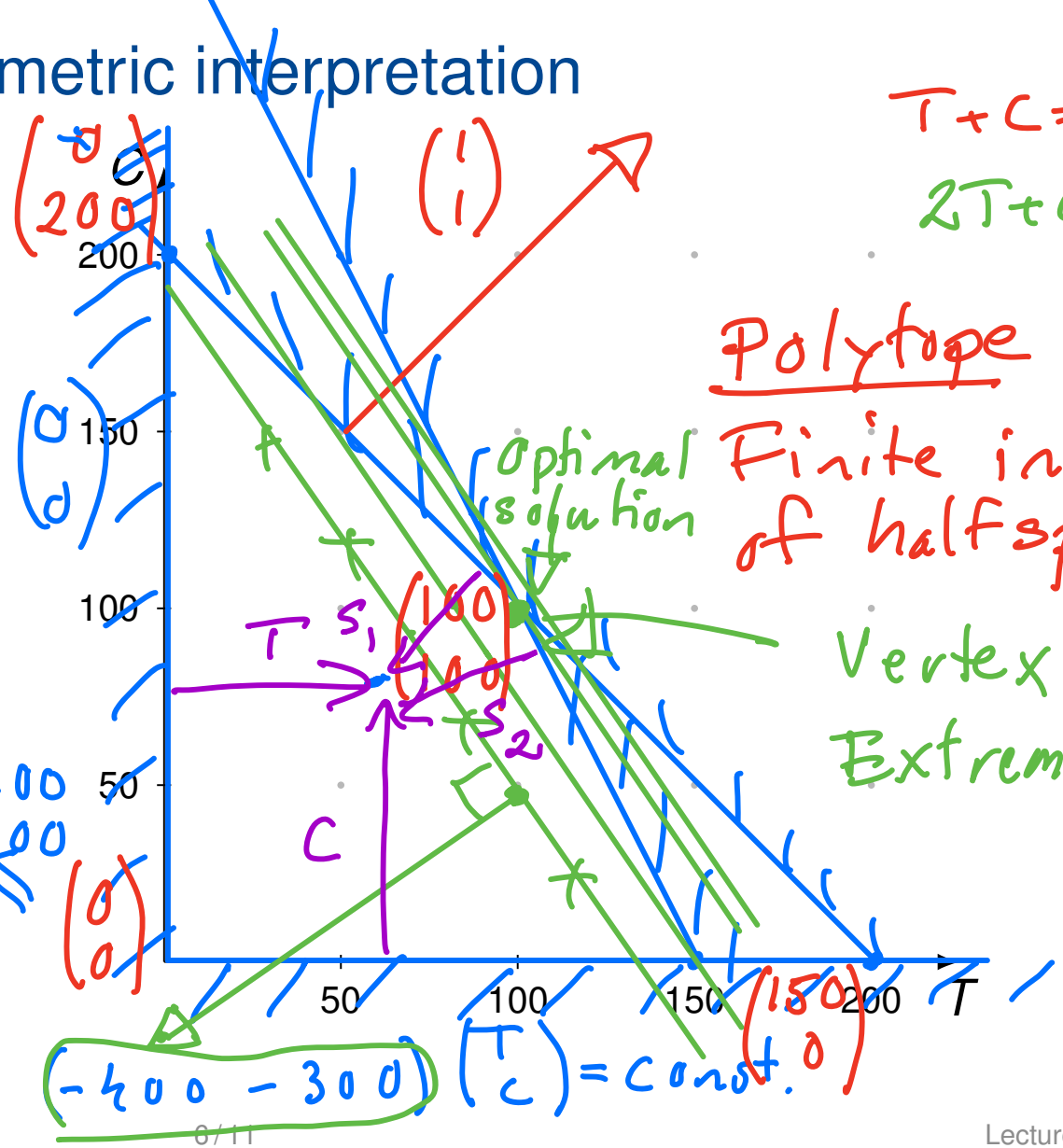
Objective function:

$\max \quad 400T + 300C.$

$\Updownarrow$

$\min \quad -400T - 300C$

$-400T - 300C = \text{const.}$



Linear programming in standard form

Component form:

$$\begin{aligned} \text{minimize} \quad & \sum_{j=1}^{n} c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^{n} a_{ij} x_j = b_i, \quad i = 1, \ldots, m, \\ & x_j \geq 0, \qquad\qquad j = 1, \ldots, n. \end{aligned}$$

m constraints
n variables
"Dimension": n − m

Vector form:

$$\begin{aligned} \text{minimize} \quad & c^{\mathsf{T}} x \\ \text{subject to} \quad & Ax = b, \\ & x \geq 0. \end{aligned}$$

$c \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$

This form is referred to as standard form.

Any linear program can be converted to standard form.

Slack variables

An inequality

$$a_i^T x \leq b_i$$

can be written as

$$a_i^T x + s_i = b_i, \qquad s_i \geq 0.$$

Hence,

$$s_i = b_i - a_i^T x.$$

The slack variable s_i measures the remaining margin in constraint i.

$T + C \leq 200$

$T + C + s_1 = 200$

$s_1 \geq 0$

$s_1 = 200 - T - C \geq 0$

$\Updownarrow$

$T + C \leq 200$

x_i not sign restricted $\Leftrightarrow$ $x_i = x_i^+ - x_i^-$, $x_i^+ \geq 0, x_i^- \geq 0$

Tables and chairs in standard form

Introduce slack variables s_1 and s_2.

$$\begin{aligned} \min \quad & -400T - 300C \\ \text{s.t.} \quad & T + C + s_1 = 200 \\ & 2T + C + s_2 = 300 \\ & T \geq 0,\ C \geq 0,\ s_1 \geq 0,\ s_2 \geq 0 \end{aligned}$$

$T \Leftrightarrow x_1$

$C \Leftrightarrow x_2$

$$\min \underbrace{\begin{pmatrix} -400 & -300 & 0 & 0 \end{pmatrix}}_{c^T} \begin{pmatrix} T \\ C \\ s_1 \\ s_2 \end{pmatrix}$$

$$\text{s.t.} \underbrace{\begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}}_{A} \begin{pmatrix} T \\ C \\ s_1 \\ s_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 200 \\ 300 \end{pmatrix}}_{b}$$

$$\begin{pmatrix} T \\ C \\ s_1 \\ s_2 \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

The maximization problem is written as a minimization problem by changing the sign of the objective function.

From geometry to algebra

At different vertices, different original variables or slack variables are equal to zero.

Vertex (T, C)	Variables equal to zero
$(0, 0)$	T, C
$(0, 200)$	T, s_1
$(100, 100)$	s_1, s_2
$(150, 0)$	C, s_2

Can the vertices be generated algebraically by choosing which variables to set equal to zero?

Main ideas

After this lecture, you should be able to:

- identify decision variables, objective function and constraints;
- formulate a simple linear programming model;
- interpret a two-variable linear program geometrically;
- introduce slack variables;
- write a linear program in standard form;
- interpret a slack variable as remaining margin.

Next lecture:

Why can an optimal solution be sought among basic feasible solutions?

vertices